

**AI-Powered Mobile Platform Enhancing Public Safety and
Community Policing in Sri Lanka**

25-26J-200

Project Proposal Report
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B.Sc. (Hons) Degree in Information Technology specialized in
Information Technology.

Department of Information Technology

Sri Lanka Institute of Information Technology
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Emergency SOS with Real-Time Location & Live Streaming
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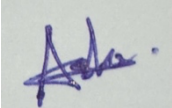
Department of Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Sri Lanka faces serious challenges in emergency communication, as existing systems such as the 119 hotline rely mainly on voice reporting, which often fails during stressful or life-threatening situations. Victims struggle to clearly explain their condition, provide accurate location details, or may be unable to speak at all, leading to delays and ineffective responses. To overcome these limitations, this research proposes a mobile-based Emergency SOS system that enables users to instantly share their real-time GPS location along with live video and audio streams directly to the nearest police or emergency unit. The solution integrates artificial intelligence (AI) to automatically analyze the live stream using YOLOv8 for detecting threats such as weapons, fire, smoke, or injuries, and YAMNet for recognizing distress sounds like shouting, alarms, or gunshots. A risk score is generated using historical crime data, helping responders prioritize cases from high-risk areas. Additionally, all geo-tagged evidence is securely stored in encrypted cloud storage to ensure accountability and transparency. By combining GPS, live streaming, AI analysis, and secure evidence management, the proposed SOS system ensures faster, smarter, and more reliable emergency responses, ultimately contributing to a safer and citizen-centered public safety framework in Sri Lanka.

Keywords: Emergency SOS, GPS Tracking, Live Streaming, Artificial Intelligence, YOLOv8, YAMNet, Real-time Emergency Response, Public Safety

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1. INTRODUCTION

Ensuring personal safety and providing timely emergency assistance have become increasingly important in modern society. Every year, thousands of people around the world face situations such as violent crimes, road accidents, natural disasters, and sudden medical emergencies, where rapid intervention methods are often limited and fail to provide Enough situational awareness to first responders. In Sri Lanka, the existing emergency hotlines like 119 the existing emergency hotlines, such as 119, rely mostly on voice communications, which depend heavily on the victim's ability to speak coherently under stress. Individuals cannot convey sufficient details owing to panic, injury, or environmental limits, resulting in delays, miscommunication, and poor replies [1].

Most existing SOS applications only provide basic data such as the caller's location or a short pre-configured text message. While GPS coordinates are useful for identifying the victim's location, they lack critical context regarding the severity of the emergency. For example, two distress warnings from the same place may appear identical, but one may be a serious armed robbery and the other a simple scuffle. Without visual or auditory context, law enforcement cannot discriminate between these circumstances, resulting in poor resource allocation and, in some cases, tragic outcomes [2]. This constraint highlights the critical need for a next-generation emergency communication system that includes more than just location information.

The planned Emergency SOS with Real-Time Location and Live Streaming technology intends to change how emergency occurrences are reported and addressed. With a single tap, users may send their GPS location, as well as a live video and audio stream, to the nearest police or emergency unit. This guarantees that authorities have immediate situational information, allowing them to assess the severity of the incident before getting on scene. The live streaming capability provides vital visual and audio information, which aids decision-making and resource allocation.

This solution is unusual in that it integrates with artificial intelligence (AI) for automatic live stream analysis. Using cutting-edge models like YOLOv8, the system can detect harmful factors including firearms, knives, fire, smoke, and visual injuries in real time [3]. At the same time, YAMNet audio categorization technology analyzes live audio feeds to detect distressing sounds such as shouting, weeping, alarms, or gunshots [4]. These AI-powered features act as intelligent filters, ensuring that genuine emergencies are prioritized while reducing false alarms. By incorporating historical crime data, the system also assigns a risk score based on the area where the incident occurs, helping authorities prioritize cases from high-risk zones more effectively [5].

Sri Lanka, like many other developing nations, faces challenges in strengthening public safety infrastructure. Police hotlines often experience high call volumes, which slow down response times, and language barriers sometimes prevent citizens from clearly explaining their emergencies. Furthermore, in severe situations such as domestic abuse, victims may be unable to talk freely for fear of escalating the threat [6]. The suggested SOS module tackles these restrictions by focusing on nonverbal and real-time digital communication, reducing the need for traditional phone calls.

Globally, governments have begun to implement mobile safety platforms. India's 112 Emergency Response Support System and the United States' E911 service are moves toward improving emergency communication. However, these solutions are primarily text- or voice-based, lacking powerful AI-powered live analysis [7]. Multimodal emergency systems, which combine location, video, and audio, outperform single-mode systems in terms of detecting the nature and severity of crises [8]. This research complements these worldwide breakthroughs while also introducing a customized, AI-enhanced model adapted to Sri Lanka's setting, making it more effective for real-world implementation.

The potential applications of this SOS module go far beyond conventional policing. During natural disasters such as floods, landslides, and fires, affected people can utilize the system to stream real-time footage of the situation, allowing for faster rescue planning.

During a car accident, victims or spectators can immediately share crash scenes, allowing emergency services to provide the necessary medical support before coming. Similarly, in the event of a medical emergency, the live video can help paramedics remotely examine the patient's status. This function provides a subtle yet powerful option for women to seek urgent help when harassed or assaulted [6].

The significance of this research stems from its capacity to integrate several technologies—GPS, live streaming, AI analysis, and crime data integration—into a single unified emergency response platform. Unlike traditional systems, which supply limited or delayed information, this technique gives authorities real-time, contextual, and validated evidence. This not only speeds up reaction times but also ensures better case prioritizing, resulting in more lives saved. Furthermore, the use of geo-tagged media evidence improves accountability and openness in police enforcement, promoting public trust.

The proposed SOS solution extends beyond immediate crisis response to lay the groundwork for a data-driven public safety ecosystem in Sri Lanka. By regularly evaluating patterns in SOS alerts and crime-prone locations, law enforcement can allocate resources to high-risk areas. This predictive power has the potential to reduce crime rates over time while also making communities safer.

Finally, this research proposal emphasizes the importance of new technology in transforming emergency communication and response. The suggested system is a significant improvement over existing SOS methods since it combines deep learning, computer vision, and audio identification into a user-friendly smartphone application. Just as automated detection technologies are altering agriculture and healthcare, AI-powered emergency systems are ready to reshape public safety. This adaptable and intelligent SOS module not only helps people in crisis, but it also improves the operational performance of law enforcement authorities. The system lays the way for a safer society by responding faster, smarter, and more evidence-based, and it serves as a model for future emergency communication frameworks in Sri Lanka and elsewhere.

1.1 Background & Literature survey

Emergency response in Sri Lanka is mostly based on hotlines like 119, which rely significantly on voice communication between victims and responders. However, in high-stress or life-threatening situations, victims frequently struggle to communicate critical data properly, resulting in delays, misinterpretation, and incorrect responses [9]. To confirm these limitations, a survey was conducted, and the results clearly indicate the challenges faced by citizens when reporting emergencies, as shown Fig. 1.1 below.

What challenges do victims face when reporting emergencies? (Select all that apply)

31 responses

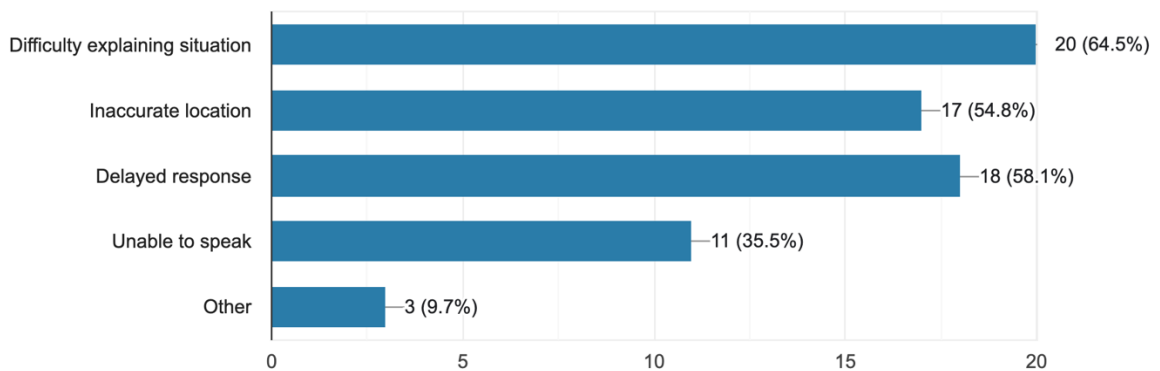


Fig 1.1: Summary of challenges do victims face when reporting emergencies.

The survey revealed that the majority of respondents (64.5%) experienced difficulty explaining their emergency situation, demonstrating how panic, stress, or injury can impair communication during crises. Similarly, 58.1% of participants reported delays in receiving responses from emergency services, underlining the inadequacies of hotline-based systems that rely primarily on verbal explanations [10]. Furthermore, 54.8% of respondents reported erroneous location reporting, which might significantly impede emergency units when time-critical action is necessary [11]. Another notable conclusion was that 35.5% of individuals reported being unable to communicate during an emergency owing to injury, environmental circumstances, or threats, while 9.7% cited other barriers

such as language difficulty or fear of retaliation [12]. These findings highlight the critical need for nonverbal, intelligent SOS systems that reduce reliance on voice communication while providing faster, more accurate emergency responses.

Similar approaches are already being adopted globally. In Europe, Next Generation 112 (NG112) allows for multimodal emergency communication, including text, video, and geolocation, which improves accessibility and real-time situational awareness [13]. In Australia, BluLink enables residents to exchange live video and GPS data with dispatchers, which improves reaction time during accidents and crimes [14]. Commercial technologies such as Carbyne offer real-time video, audio, and location-sharing capabilities while lowering response times by more than 60% in pilot locations [15]. Furthermore, current artificial intelligence research has shown that models like YOLOv8 can detect firearms, smoke, or injuries in real-time video streams, whilst YAMNet can accurately distinguish distress sounds like shouting, alarms, or gunshots [16], [17]. Multimodal techniques that include visual and audio inputs regularly beat single-modality algorithms for identifying violence and prioritizing emergencies [18].

Therefore, the proposed SOS solution directly addresses the communication challenges highlighted in the survey (Fig. 1.1) by integrating GPS tracking, live video and audio streaming, AI-powered threat detection, and secure evidence storage into a single platform. This multimodal system ensures that emergency responders receive not only accurate location details but also real-time situational awareness, enabling faster and more effective decision-making. By moving beyond outdated voice-only hotlines, Sri Lanka can establish a modern, AI-enhanced emergency communication framework that is more reliable, transparent, and citizen-centered.

If yes, how did you seek help?

31 responses

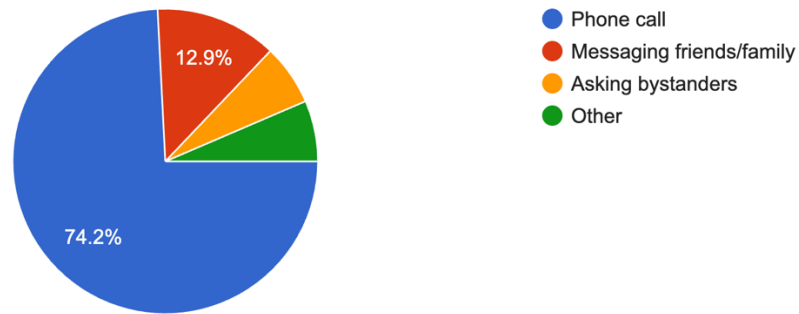


Fig 1.2: Summary of helping seeking methods.

The survey also examined how individuals sought assistance during emergencies. As illustrated in Figure 1.2, the majority of respondents (74.2%) relied on making a phone call, usually to the police hotline or family members. A smaller proportion (12.9%) reported messaging friends or family, while others sought help by asking bystanders (8.1%) or through other means (4.8%). These results clearly indicate that citizens in Sri Lanka remain heavily dependent on voice calls as their primary method of communication during emergencies [19]. However, as highlighted in earlier findings, reliance on phone calls is problematic because victims often face difficulty explaining the situation, providing accurate location details, or may even be unable to speak in critical circumstances [10], [12]. This dependency exposes the limitations of the existing emergency framework and underscores the necessity for a modern SOS solution that integrates real-time video, audio, and automated GPS location sharing, thereby reducing the risks associated with miscommunication and delayed responses [13], [14].

Did you feel the help you received was timely and effective?

31 responses

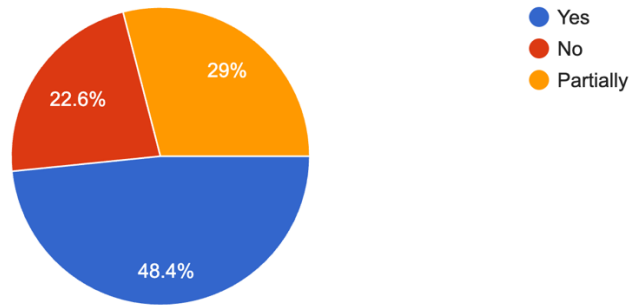


Figure 1.3: Summary of responses for People are not aware of separating these symptoms.

Previous studies have research has consistently shown that the efficiency of emergency services is determined not just by the availability of hotlines, but also by their capacity to provide timely and correct responses. In Sri Lanka, hotline-based systems like 119 have been criticized for being slow, ineffective, and excessively reliant on victims' capacity to communicate vocally [9], [10]. These limitations are displayed in the survey conducted for this study: as shown in Figure 1.4, only 48.4% of respondents believed that the support they received was timely and effective, while 29% thought it was only somewhat successful, and 22.6% were completely dissatisfied. This suggests that the majority of participants (51.6%) had unfavorable or mixed experiences, implying that most individuals do not have a positive opinion of the effectiveness of present emergency services. This finding underlines the inconsistency of existing systems and reinforces global research emphasizing the necessity of dependability and responsiveness in disaster management [11].

Academic research confirms that response quick is one of the most important variables in saving lives during an emergency. According to Panagiotopoulos et al. [19], digital crisis communication can improve real-time coordination between citizens and government authorities, resulting in faster resource dispatch. Similarly, Giannakopoulos et al. [20] demonstrated that multimodal analysis (video and audio) enhances decision-making by providing responders with contextual data that greatly outperforms voice-only communication. These findings are consistent with our survey results, in which

respondents indicated unhappiness with current phone-based systems and emphasized the need for more evidence-based, real-time solutions.

Globally, governments have begun to move to next-generation emergency systems. The Next Generation 112 (NG112) effort in Europe incorporates multimedia elements such as video calls, geolocation, and instant messaging with standard emergency numbers, giving responders a more complete picture of the situation [13]. Australia's BluLink platform allows callers to transmit live video and exchange GPS coordinates with dispatchers, dramatically reducing confusion and enhancing accuracy [14]. In the private sector, Carbyne has shown that cloud-based emergency systems that include real-time video, audio, and geolocation can reduce response times by up to 65% [15].

Artificial intelligence has also emerged as an important technique for improving accuracy and timeliness. Redmon et al. [22] shown that YOLOv8 can detect dangers such as weapons or fire in real time, enabling for more efficient event prioritization. Meanwhile, Google's YAMNet audio model can distinguish distress-related sounds like weeping, alarms, and gunshots, allowing responders to better determine urgency [21]. These solutions immediately address the discontent shown in Figure 1.4 by decreasing the need for manual reporting and enabling automated, AI-driven emergency verification.

The survey findings combined with prior research suggest that the main weakness of existing systems is not just accessibility but also trust and consistency in outcomes. If citizens cannot rely on emergency services to respond effectively, confidence in the system erodes. By integrating GPS tracking, live multimedia streaming, and AI-powered analysis, the proposed SOS solution has the potential to overcome these limitations, ensuring faster, more consistent, and more trustworthy emergency responses for Sri Lanka.

Research Gap

Emergency response systems have been investigated for more than a decade, however the majority of existing solutions still have limitations in terms of providing real-time intelligence. Current SOS apps are generally limited to providing short SMS alerts or GPS locations, which provide no situational information for responders. Kumar and Singh [23] suggested a mobile-based emergency response solution that used GPS tracking, however the system only transmitted static coordinates and no contextual or multimedia data. Due to a lack of contextual information, responders are unable to determine the seriousness of an incident until they arrive on the site, which frequently results in delays or inappropriate resource deployment. Thus, there is a huge gap in SOS systems' ability to provide comprehensive, real-time insight in addition to location data.

Table 1. 1: Comparison between existing systems

Application Reference	Real-time GPS Location Sharing	Live Video/Audio Streaming	AI-Powered Incident Analysis (YOLOv8, YAMNet)	Risk-based Prioritization (Crime Data)	Geo-tagged Evidence Validation	Multilingual Support (Sinhala/English)
Research A	✓	✗	✗	✗	✗	✗
Research B	✓	✓	✗	✗	✗	✗
Research C	✓	✓	✗	✗	✗	✗
Proposed System	✓	✓	✓	✓	✓	✓

Research A focuses on the development of GPS-enabled SOS applications where users can share their real-time geographic location with emergency contacts or authorities using mobile devices [24]. The strength of this approach lies in its simplicity, as it allows responders to locate individuals quickly even when they are unable to verbally describe their location. Such systems are particularly valuable in rural or unknown areas, where precise navigation is essential for timely intervention. However, these systems are highly

limited in scope, as they only provide static location data without any contextual or multimedia evidence. This lack of live information means responders cannot evaluate the seriousness of the situation until they arrive at the location, which often delays the deployment of appropriate resources. Furthermore, these systems do not integrate AI-powered threat detection or validation mechanisms, leaving them vulnerable to false alarms and misuse. As Majumdar and Awasthi [27] argue, relying solely on GPS also carries risks of spoofing or manipulation, which reduces trust in such applications. Compared to the proposed system, Research A highlights the importance of real-time location sharing but remains a single-modality solution that fails to deliver situational awareness or automated incident verification.

Research B builds upon traditional SOS systems by combining GPS tracking with WebRTC-based live video streaming, allowing responders to observe emergencies in real time [22]. This innovation provides significant improvements in situational awareness because authorities can view the actual context of an event rather than relying solely on victim descriptions. By monitoring the live feed, responders are able to distinguish between genuine emergencies and false alerts, which enhances efficiency. However, despite these improvements, the system still suffers from limitations. The interpretation of live video streams depends entirely on human operators, which is impractical when there are multiple simultaneous alerts. Moreover, the system does not include AI-powered models such as YOLOv8 for weapon or fire detection or YAMNet for analyzing audio streams for distress-related sounds. Without such automation, the platform remains reactive, with authorities responsible for manually analyzing every case. In addition, it lacks geo-tagged evidence validation and risk-based prioritization, meaning all alerts are treated equally regardless of urgency or context. Compared to the proposed system, Research B advances the field by introducing live video streaming but remains dependent on manual monitoring, making it less efficient and scalable.

Research C, on the other hand, is centered on women’s safety applications that integrate GPS tracking with the option to send SOS alerts and sometimes live video or audio streams to trusted contacts or law enforcement [24]. These systems are designed with accessibility in mind, ensuring that vulnerable populations can quickly alert authorities in unsafe

situations. The combination of location and media improves user confidence and often results in quicker response times. While this research improves inclusivity, it too demonstrates several gaps. The system lacks AI-driven analysis that could automatically detect threats such as violence or hazardous environments, which limits its ability to distinguish between critical and non-critical cases. It also fails to integrate risk scoring based on crime-prone areas or contextual intelligence, leading to inefficient resource allocation. Another limitation is the absence of backend verification dashboards that would enable authorities to filter prank or false alerts, a challenge highlighted in citizen-reporting systems [28]. Furthermore, there is no multilingual support, which restricts usability in regions like Sri Lanka where Sinhala, Tamil, and English coexist as primary languages. Thus, although Research C highlights the importance of accessibility and user-friendliness, it reflects a fragmented approach lacking automation, prioritization, and localization.

In contrast to these prior studies, the proposed system introduces a comprehensive and intelligent emergency SOS solution that integrates multiple features into a single platform. It combines real-time GPS tracking with live video and audio streaming, enabling responders to not only know the victim's location but also observe the incident as it unfolds. To overcome the limitation of human-only interpretation, the system incorporates AI-powered analysis using YOLOv8 to detect weapons, fire, or smoke in video feeds, and YAMNet to recognize distress-related sounds such as shouting, alarms, or gunshots. Unlike previous approaches, it further introduces risk-based prioritization, which combines historical crime data with live AI-detected threats to rank emergencies according to severity. This allows law enforcement to allocate resources more effectively and address the most critical cases first. Additionally, the system validates multimedia alerts through geo-tagged evidence, reducing false alarms and prank alerts. Finally, by providing multilingual support in Sinhala, Tamil, and English, it ensures inclusivity and accessibility for diverse users in Sri Lanka. Collectively, these features address the limitations of Research A, B, and C, transforming the proposed system into a holistic, intelligent, and context-aware platform that ensures faster police response, accurate incident verification, and improved citizen trust in digital safety technologies.

1.2 Research Problem

Current emergency response systems lack the intelligence and multimodal integration required to provide effective real-time assistance. Existing solutions such as GPS-enabled SOS applications provide only static location information, leaving responders without situational awareness of the incident [2]. While useful in narrowing down victim location, these systems are limited by their single-modality design and are vulnerable to false alerts and GPS spoofing, which reduces their reliability and effectiveness in high-risk scenarios [5].

More advanced systems that integrate live video streaming with geolocation tracking have been proposed [3]. Although these approaches provide richer contextual information than static GPS data, they remain highly dependent on human operators to interpret live streams. This manual process is inefficient in large-scale crises and does not scale well when multiple alerts are received simultaneously. Furthermore, these systems fail to incorporate AI-powered detection of threats such as weapons, fire, smoke, or distress-related sounds, leaving critical incidents unclassified and delaying the prioritization of urgent cases [6].

Applications focusing on women’s safety combine GPS with multimedia alerts, offering faster and more accessible emergency communication [1]. However, these systems still lack backend verification mechanisms to filter prank or false alerts, which undermines trust and reduces their adoption by law enforcement [8]. They also fail to provide risk-based prioritization, treating all alerts as equally urgent regardless of context, and lack multilingual support, limiting their usability in diverse populations such as Sri Lanka where Sinhala, Tamil, and English coexist as primary languages.

Therefore, the central research problem is the absence of an intelligent, multimodal, and context-aware SOS system that integrates real-time GPS tracking, live audio and video streaming, AI-powered threat detection, risk-based prioritization, and geo-tagged evidence validation into a single platform. Without such a system, emergency responders face delays, inefficiencies, and high error rates in identifying, verifying, and prioritizing

incidents, which significantly undermines the speed and effectiveness of emergency interventions.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this research is to develop an AI-powered Emergency SOS system that integrates real-time GPS tracking, live audio/video streaming, and automated threat analysis to provide faster, more accurate, and effective emergency response in Sri Lanka.

2.2 Specific Objective

- To design a one-touch SOS feature that instantly shares a user's live location with the nearest police or emergency unit.
- To implement real-time video and audio streaming so responders can observe the situation directly.
- To integrate YOLOv8-based computer vision for detecting critical threats (e.g., firearms, fire, smoke, visible injuries) in live streams.
- To apply YAMNet-based audio analysis for recognizing distress sounds such as shouting, crying, alarms, or gunshots.
- To incorporate geo-tagged media evidence and risk scoring (using historical crime data) for prioritizing high-risk alerts.
- To develop a secure backend and police dashboard for validated alerts, displaying live feeds, locations, and severity scores to support decision-making.
- To ensure data security and evidence preservation by storing emergency video/audio in secure cloud storage.

3. METHODOLOGY

3.1 Understanding the key pillars of the research domain.

3.1.1 Real-Time GPS Tracking

Accurate geolocation is an essential component of any emergency response system. The SOS module will use the smartphone's GPS sensor to send the user's current coordinates. Unlike traditional voice reporting, GPS provides responders with exact positional data without relying on the victim's ability to identify their location [1]. Continuous tracking during an active alert enables responders to track movements (for example, in kidnappings or moving cars), eliminating communication delays.

3.1.2 Live Audio/Video Streaming

The proposed solution uses the smartphone's camera and microphone to offer real-time multimedia evidence. When the SOS button is pressed, the device sends live video and audio to the police dashboard, allowing rescuers to remotely monitor the situation before arriving on location. This enhances situational awareness and aids in the differentiation of low- and high-severity cases, addressing one of the major shortcomings of Sri Lanka's present hotline (119), which lacks contextual information [2].

3.1.3 Computer Vision with YOLOv8

The video feed will be processed by YOLOv8, an advanced deep learning model for object detection. YOLOv8 detects firearms, knives, fire, smoke, and visual injuries in real time [16]. This automation lowers the need for human interpretation by operators and guarantees that alarms are prioritized when potentially life-threatening items are discovered. A call reporting a brawl, for example, may be instantly escalated if YOLOv8 detects a weapon in the live stream.

3.1.4 Audio Event Detection with YAMNet

When visual input is covered (for example, due to low light or a blocked camera), audio is as important. The system will analyze sound using YAMNet, an audio classification model that can identify events like screaming, sobbing, alarms, and gunshots [4]. These cues supplement visual evidence, allowing emergency personnel to better assess the seriousness of the

occurrence. Combining both senses improves reliability, as demonstrated by multimodal violence detection study [8].

3.1.5 Risk Scoring and Geo-Tagging

Beyond live evidence, the SOS module will incorporate historical crime data to assess the risk level of the incident location. By referencing crime-prone zones, the system assigns a “risk score” that guides prioritization. For example, two alerts from different areas may appear identical, but the system will prioritize the one occurring in a known high-crime zone [19]. This risk-aware approach ensures limited resources are allocated to the most urgent cases.

3.1.6 Secure Backend and Police Dashboard

All data transmitted from the SOS application will be routed through a secure backend service. The backend performs three key tasks:

1. Validation – Verifying alerts to reduce false alarms.
2. Storage – Encrypting and storing geo-tagged evidence (video, audio, location) for accountability and legal purposes.
3. Distribution – Delivering validated alerts to the police dashboard, which displays real-time video feeds, GPS tracking, AI-detected threats, and risk scores.

The dashboard ensures that law enforcement officers receive actionable intelligence rather than fragmented reports, improving both response time and operational transparency. Blockchain-based storage may also be explored to guarantee evidence integrity, similar to models applied in digital policing solutions [7].

3.2 System Architecture Diagram

Inside of the component there are several tasks which are divided for the purpose of understanding and to do further the implementations.

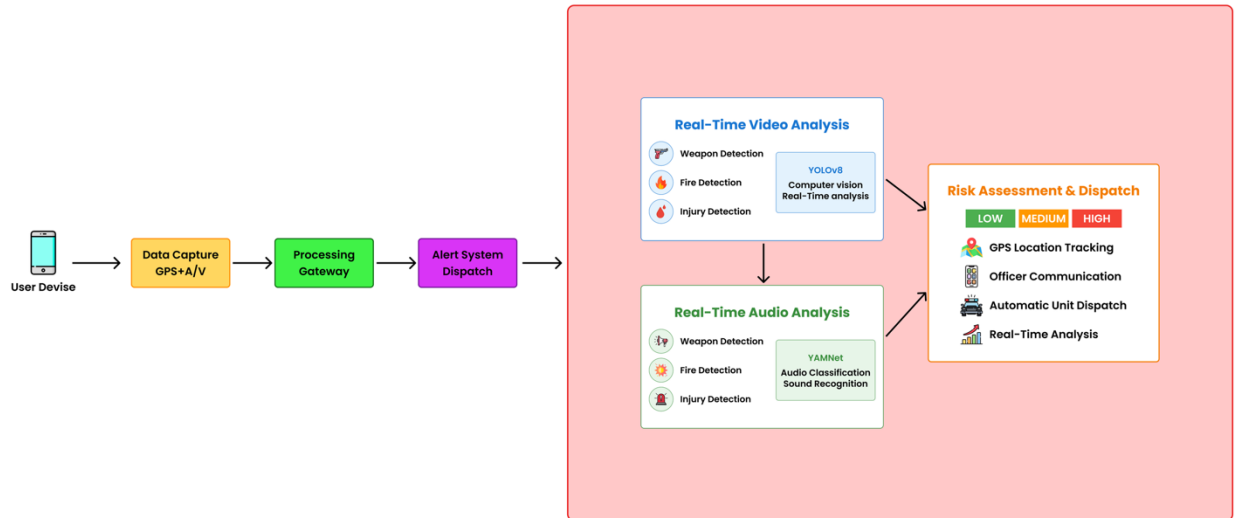


Figure 3. 3: High Level Architecture Diagram

The proposed Emergency SOS with Real-Time Location and Live Streaming component is designed to provide rapid, reliable, and intelligent emergency communication between citizens and law enforcement. The process begins when a user activates the SOS button within the mobile application. This single action triggers multiple processes at once: the system starts GPS tracking to record the victim's exact location, while the smartphone camera and microphone are activated to capture live video and audio streams. These inputs are immediately packaged with time stamps and session identifiers and securely transmitted to the backend system. Unlike traditional voice-based emergency calls, this process reduces the dependency on the victim's ability to speak clearly during high-stress situations.

At the backend, the SOS alert undergoes a validation and authentication process to ensure that the data is accurate and securely transmitted. Once validated, the live streams and location data are routed to two different pipelines: the AI analysis services and the Police Dashboard interface. The Police Dashboard provides responders with instant access to the caller's GPS coordinates, a live video/audio feed, and contextual metadata. This ensures that

responders can view what is happening at the scene even before they arrive, leading to faster and more accurate situational awareness.

The system also incorporates artificial intelligence to analyze the live feed in real time. The video stream is processed using YOLOv8, an advanced object detection model, which is trained to detect critical risk indicators such as firearms, knives, fire, smoke, or visible injuries. At the same time, the audio stream is analyzed by YAMNet, which recognizes distress-related sounds such as shouting, crying, alarms, or gunshots. These detection services generate structured results with confidence scores that describe the likelihood of a detected threat. By relying on AI-driven analysis, the system ensures that false alarms are minimized, and genuine emergencies are prioritized.

Especially, The outputs from video and audio analysis are combined in a severity scoring module. This module fuses the AI detection confidence levels with contextual information such as the victim's environment and historical crime data of the location. For example, two similar alerts may occur at different locations, but the system will prioritize the one originating from a known high-crime area. The final output is a risk score, which classifies emergencies into categories such as low, medium, or high severity. This scoring process allows police officers and dispatchers to focus their resources on the most urgent situations, reducing response times for life-threatening cases.

Once the severity score is determined, the information is displayed on the Police Dashboard, which provides an interactive interface for emergency responders. Here, officers can view the live video and audio streams, track the user's precise GPS location, and monitor the automatically calculated severity score. Responders also have the ability to communicate with the victim if safe to do so, or to silently monitor the situation if direct communication might escalate the threat. Alongside the dashboard, the backend system issues dispatch notifications via SMS, email, or push alerts to the nearest available units, ensuring that emergency services are mobilized immediately.

Finally, every SOS alert is accompanied by **secure evidence storage**. The system automatically encrypts and saves geo-tagged media (video, audio, and location data) to

secure cloud storage. This evidence is time-stamped and can later be used for investigations or court proceedings, ensuring transparency and accountability. Even if the victim's device is damaged or disconnected, buffered media is uploaded once the connection is restored. This ensures that critical evidence is never lost, and victims can rely on the system to protect their data and safety.

3.3 Project Requirements

3.3.1 Functional Requirements and Non-Functional Requirements

Table 3. 1: Functional and Non-functional requirements

Functional Requirements	Non-Functional Requirements
The system should allow one-touch SOS activation to immediately trigger an alert.	user interface should be simple and user-friendly
The system should capture and share real-time GPS location of the user with the nearest police or emergency unit.	function reliably across platforms
The mobile app must enable live video and audio streaming during emergencies for direct observation by responders.	low-latency performance
The system should validate the SOS alert and ensure that only accurate, authenticated data is forwarded to authorities.	high reliability and availability
The application should analyze the live stream using AI models (YOLOv8 and YAMNet) to detect weapons, fire, smoke, distress sounds, or visible injuries.	high accuracy
The system should assign a risk score to incidents based on historical crime data and the detected severity level.	efficient
The backend should store geo-tagged video/audio evidence securely, ensuring that data can be used later for accountability and legal purposes.	fast response times

The police dashboard should display validated alerts, live feeds, GPS location, and risk scores to support decision-making.	high user experience
The system should enable two-way communication between the victim and responders if required.	strict security and privacy protocols

3.3.2 User Requirements

- A user should be able to trigger the SOS alert with a single tap on the emergency button.
- A user should be able to share their real-time GPS location automatically with responders.
- A user should be able to stream live video and audio directly from their device to the police/emergency dashboard.
- A user should be able to communicate with responders through two-way communication (optional voice or text).
- A user should be able to receive confirmation that their SOS request has been received by authorities.
- A user should be able to view nearby police stations or emergency units within the app.
- A user should be able to store emergency contacts (family/friends) who are notified when an SOS alert is triggered.
- A user should be able to operate the app offline, with the alert automatically sent once connectivity is restored.
- A user should be able to trust that their data (video, audio, GPS) is securely stored and only accessible by authorized responders.
- A user should be able to access the application in multiple languages (Sinhala, English).

3.3.3 System Requirements

- Software requirements

Flutter

Python

FastAPI

React / React Native

TensorFlow

Google Map

MySQL / MongoDB

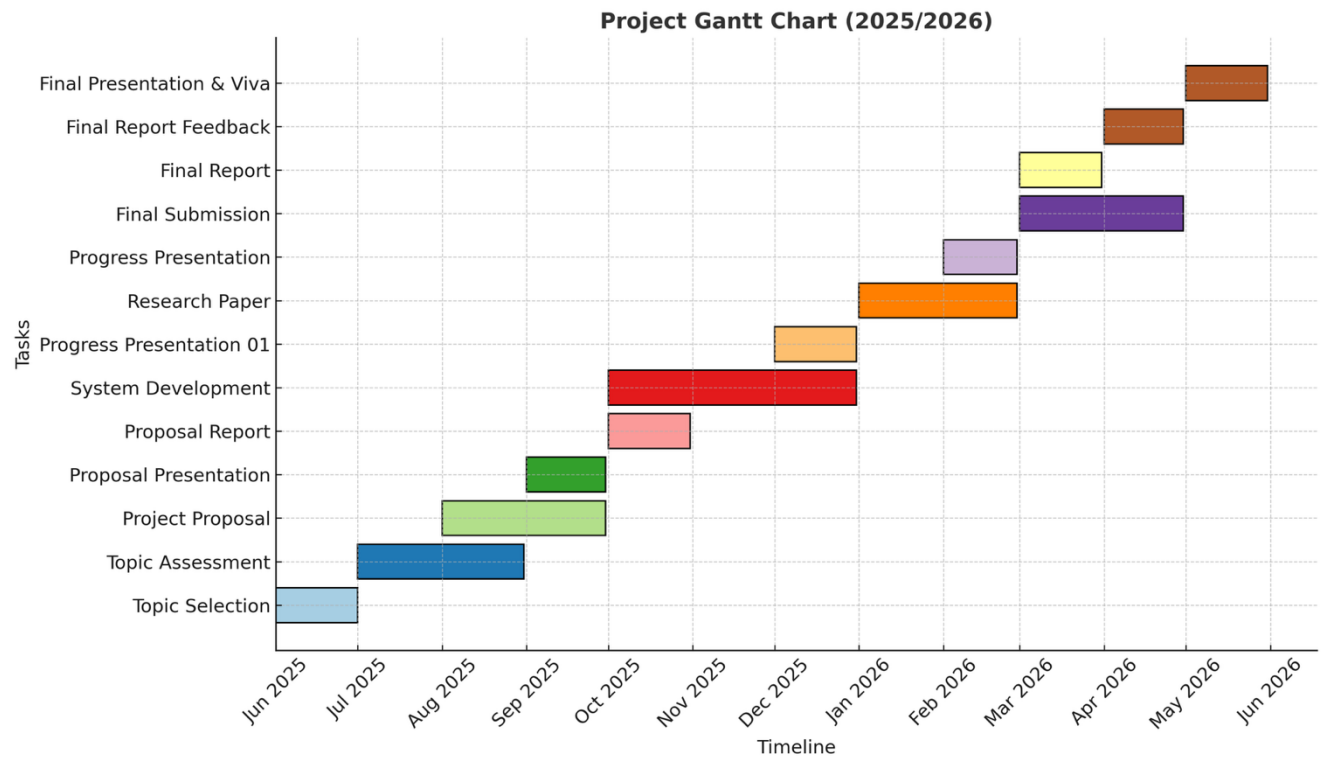
AWS

3.3.4 Personnel Requirements

Collecting resources and dataset of diseases.

- Sri Lanka Police
- Moderators

4. GANTT CHART



5. BUDGET AND BUDGET JUSTIFICATION

Table 6. 1: Budget Plan

Requirement	Cost (Rs.)
Travelling Cost for data collection	6000.00
vps	7500
Mobile app hosting on Play Store	8200.00
Mobile app hosting on App Store	20000
Total	39700

6. COMMERCIALIZATION

6.1 Target Audience and Market Space.

- **Target Audience.**

Sri Lankan citizens.

Researchers and citrus Research Institute.

- **Market Space.**

No need for advanced knowledge in technology.

No age limitation for the users.

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APPENDICES

Survey link

https://docs.google.com/forms/d/e/1FAIpQLSfrnohFy4FI_bKuHuiGacYFRRrP5EtYcUeiMNpDzsmB13ji3Q/viewform?usp=dialog